

Karthik Reddy Yalala

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EDUCATION

Arizona State University

Tempe, AZ

Bachelor's in Computer Science (GPA: 4.0)

May 2027

- Relevant Coursework: Algorithms, Data Structures, Object-Oriented Programming, Operating Systems, Software Engineering, Cybersecurity, Artificial Intelligence, Machine Learning, Web Development, Discrete Mathematics
- Dean's List: Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2026

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, HTML/CSS, Python, Java, C/C++

AI/ML & Agentic Tools: Keras, TensorFlow, NumPy, Pandas, Matplotlib, Hugging Face Transformers, GPT (Prompt Engineering), LangChain, CrewAI, RAG

Web Development: React.js, Node.js, Express.js, REST APIs, Tailwind CSS, Streamlit, Responsive Design

Databases: PostgreSQL, MySQL, MongoDB, Firebase, SQL

Cloud & DevOps: AWS (EC2, S3, IAM), Docker, Git, GitHub, CI/CD (basic), Linux, Vercel, Cloudflare

CERTIFICATIONS

AWS Certified Cloud Practitioner (CLF-C02)

PROFESSIONAL EXPERIENCE

Software Engineer Intern

May 2024 – July 2024

YJR Realtors

Hyderabad, India

- Developed a full-stack web application using React.js and Node.js, designing RESTful APIs to manage property listings and customer inquiries
- Implemented JWT-based authentication and secure backend workflows, integrating MongoDB for lead tracking and data persistence
- Optimized database queries and automated follow-up systems, reducing average response time by 40%

Data Analyst Intern

May 2025 – July 2025

Food Forest AI

Philadelphia, PA

- Built Python-based data preprocessing pipelines to clean, validate, and standardize 200+ company records for system integration
- Engineered web scraping and structured parsing pipelines to extract normalized product and service metadata from HTML sources
- Implemented deduplication logic and anomaly detection workflows to improve dataset accuracy by 40%

TECHNICAL PROJECTS & ACHIEVEMENTS

Multi-Agent Research Automation System | CrewAI, LangGraph, OpenAI APIs, Pinecone

June 2025

- Architected a modular multi-agent workflow system using planner-executor architecture with Retrieval-Augmented Generation (RAG)
- Integrated Pinecone vector database and optimized API orchestration for concurrent task execution and structured result generation
- Designed robust task state management and response validation pipelines to improve reliability and reduce manual research effort by 70%

Real-Time Chat Platform | Node.js, Express, MongoDB, Socket.io, JWT

May 2024

- Designed and implemented scalable RESTful APIs and WebSocket-based real-time messaging using Socket.io with secure JWT authentication
- Architected MongoDB schema for efficient message storage and indexing, implementing rate limiting and backend security controls for production deployment
- Designed event-driven message broadcasting architecture to handle concurrent client connections with low-latency delivery